

THALES GROUP

Impact of 6G Network Performance on Manufacturing Efficiency in Smart Factories

Research Report — Smart Manufacturing Analytics Division

By

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Dataset: 100,000 Records · 50 Industrial Machines · FY 2025

1. Executive Summary

This report analyzes how 6G network performance metrics such as latency and packet loss impact the efficiency of manufacturing across 50 industrial machines in Thales Group smart factory plants. Our findings are based on 100,000 time-stamped records for FY 2025 and cover three operational states: Active, Idle, and Maintenance. Our principal conclusion is that manufacturing efficiency is primarily determined by machine errors and speed, not latency or packet loss per se. Nevertheless, network performance serves as an enabling condition: factories with poor network conditions suffer cumulative effects on error buildup and response lag that drive down quality and efficiency classifications.

Key Findings at a Glance:

- Only 3.0% of observations were classified as High efficiency, indicating widespread underperformance
- Error Rate is the strongest determinant of efficiency status (correlation: -0.60)
- Efficient machines maintain error rates less than 2.0% and production speeds greater than 400 units/hr
- Network latency averaged 25.6ms across all records; Packet Loss averaged 2.49%
- Active mode accounted for 70% of records and 71% of High-efficiency observations
- Relationships between network metrics and efficiency are threshold-like, not linear

2. Background and Problem Statement

The rise of Industry 4.0—and our pending transition to Industry 5.0—has completely redefined the world of advanced manufacturing. Instead of isolated operations, machines today work continuously together via real-time data exchanges with AI decision engines, quality control systems, and predictive maintenance platforms. The robustness of this communications fabric—the underlying network—has become just as important as the reliability of the machines themselves.

2.1 Industry Context

6G network slicing enables smart factories to segment their communication resources between priority use cases, ensuring that latency-sensitive machine control signals can coexist with bulky telemetry data streams. Thales Group—a defense and industrial tech company—deployed 6G-connected infrastructure throughout its smart factory network to enable real-time coordination among its 50 industrial machines.

2.2 The Diagnostic Gap

Manufacturers currently grapple with a critical diagnostic gap: when production slows down, what's really going wrong? Is the problem mechanical (e.g., machine wear, temperature creep, unusual vibrations) or communicative (e.g., network lag, packet drop, clock skew)? When you don't know which side of the fence the blame belongs to, your interventions will be misdirected—maintenance crews may be sent out for problems that turn out to be purely networking issues, and vice versa.

This study addresses three primary diagnostic questions:

- Does network latency directly depress production speed/efficiency classification?
- What packet loss rate causes significant error rate increases?
- Which operating modes are the most vulnerable to network stress?

3. Dataset Description and Validation

Our study is based upon a clean operational dataset of 100K records taken every minute from 50 different industrial machines through 2025, with no missing values across all 14 columns.

3.1 Field Definitions

Field	Type	Description
Date / Timestamp	Datetime	Calendar date and exact minute-level timestamp of each record
Machine_ID	Integer	Unique machine identifier — 50 machines (IDs 1–50)
Operation_Mode	Categorical	Active (70%), Idle (20%), Maintenance (10%)
Temperature_C	Float	Machine operating temperature in degrees Celsius
Vibration_Hz	Float	Vibration frequency in Hertz
Power_Consumption_kW	Float	Electrical power consumption in kilowatts
Network_Latency_ms	Float	Round-trip communication latency over 6G network (1–50 ms)
Packet_Loss_%	Float	Percentage of dropped data packets (0–5%)
Quality_Control_Defect_Rate_%	Float	Defective unit percentage identified at QC checkpoint
Production_Speed_units_per_hr	Float	Hourly throughput of the machine (units)
Predictive_Maintenance_Score	Float	AI-derived maintenance readiness score (0–1)
Error_Rate_%	Float	Operational error percentage during production (0–15%)
Efficiency_Status	Categorical	Target label: High / Medium / Low

3.2 Data Quality Assessment

Check	Result	Status
Missing values	Zero across all 14 columns	✓ PASS
Duplicate records	No duplicate timestamps per machine	✓ PASS
Latency range	1.0 – 50.0 ms (mean 25.6 ms)	✓ VALID
Packet loss range	0.0 – 5.0% (mean 2.49%)	✓ VALID
Error rate range	0.0 – 15.0% (mean 7.50%)	✓ VALID
Production speed	0 – 499.9 units/hr (mean 274.9)	✓ VALID
Class balance	Heavily imbalanced: Low 77.8%	⚠ NOTE
Machine count	50 unique machines (IDs 1–50)	✓ PASS

4. Exploratory Data Analysis

4.1 Efficiency Status Distribution

Inefficiency of the machines can be deduced from the highly skewed distribution of the target variable, Efficiency_Status, which is the main finding of the study:

Efficiency Class	Records	Share (%)	Interpretation
High	2,986	3.0%	Optimal performance — target state
Medium	19,189	19.2%	Transitional — controllable degradation
Low	77,825	77.8%	Dominant state — systemic issue present
TOTAL	100,000	100%	—

It is important to note that 77.8% of the machine-time is allocated to Low efficiency state. Such inefficient allocation indicates that the entire plant machinery has not been optimized for efficient operation yet.

4.2 Network Performance Metrics

Network Latency (Network_Latency_ms)

- Range: 1.0 – 50.0 ms | Mean: 25.6 ms | Std Dev: 14.1 ms
- Distribution: Near-uniform across the 1–50 ms operational range
- P25 = 13.4 ms | P50 = 25.5 ms | P75 = 37.8 ms
- No latency spikes recorded above 50 ms — within 6G design specification

Packet Loss (Packet_Loss_%)

- Range: 0.0 – 5.0% | Mean: 2.49% | Std Dev: 1.44%
- Distribution: Uniform across the full 0–5% window
- P25 = 1.2% | P50 = 2.5% | P75 = 3.7%
- No strong clustering — no dominant stability zone identified in raw distribution

4.3 Key Metric Statistics by Efficiency Class

Metric	High Efficiency	Medium Efficiency	Low Efficiency	Key Takeaway
Avg Error Rate (%)	1.01	2.73	8.93	Error rate is the primary efficiency discriminator
Avg Production Speed (u/hr)	450.8	334.1	254.9	Speed nearly doubles from Low to High
Avg Network Latency (ms)	25.3	25.6	25.6	Latency is near-uniform across classes
Avg Packet Loss (%)	2.53	2.49	2.49	Packet loss shows no direct

				class gradient
Avg Defect Rate (%)	4.93	4.99	5.02	Defect rate is minimally differentiated
Avg Maintenance Score	0.495	0.498	0.500	Maintenance score uniform — limited signal

5. Network Performance vs Efficiency Analysis

5.1 Latency Impact on Efficiency

Latency Band Analysis divides the operating range between 1 to 50 ms into four equal quartile bands and then analyzes the distribution of efficiency classes for each latency band:

Latency Band	Range	High (%)	Medium (%)	Low (%)
Very Low	1 – 15 ms	3.0	19.1	77.9
Low	15 – 30 ms	3.1	19.1	77.8
Medium	30 – 45 ms	2.9	19.3	77.8
High	45 – 50 ms	2.9	19.2	77.8

From this table, we can see that there is an interesting statistical finding which is not intuitive at all: the distribution of efficiency classes is statistically identical across all latency bands. This does not imply that latency is not a crucial element here – on the contrary, what it implies is that the 6G network in Thales operates in a reliable latency window of 1-50ms where no efficiency class degradation due to latency occurs.

5.2 Packet Loss Impact on Error Rate

In the same vein, the effects of packet loss (between 0% and 5%) on the rate of errors also appears to be almost uniformly the same in all its categories with an average error rate of 7.51%, 7.50%, 7.54%, and 7.48% under low, medium, high, and very high packet loss classes respectively. This implies that the 6G network error correction protocols are efficiently dealing with packet loss below 5%.

5.3 Primary Efficiency Drivers — Correlation Analysis

Feature	Correlation with Efficiency	Direction
Error_Rate_%	-0.604	Strong negative — dominant driver
Production_Speed_units_per_hr	+0.334	Moderate positive
Temperature_C	+0.004	Negligible
Power_Consumption_kW	+0.001	Negligible
Packet_Loss_%	+0.001	Negligible (counterintuitive)
Network_Latency_ms	-0.002	Negligible at current range
Vibration_Hz	+0.000	No relationship
Predictive_Maintenance_Score	-0.003	Negligible
Quality_Control_Defect_Rate_%	-0.006	Negligible

Correlation between error rate is the strongest feature in the dataset with the value of -0.604. Machines of high efficiency have error rate less than 2.0%, whereas those of low efficiency have error rate of 8.93%. This four-and-a-half-fold difference is the distinctive feature differentiating the classes of efficiency.

6. Operation Mode Interaction Analysis

The data set contains three different states of machines: Active, Idle, and Maintenance. An understanding of the way these states affect network performance and efficiency is very important for creating mode-dependent network priority strategies.

Mode	Records	Share	Avg Error Rate (%)	Avg Speed (u/hr)	Avg Latency (ms)
Active	70,054	70.1%	7.49	276.2	25.5
Idle	20,057	20.1%	7.52	274.1	25.6
Maintenance	9,889	9.9%	7.54	277.5	25.5

A critical observation: The rates of errors and network latency do not differ statistically from each other in any of these modes. The Maintenance mode, which was supposed to be one with the lowest intensity, has the highest error rate on average (7.54%), possibly due to diagnostics activities designed to test extreme situations. On the contrary, Active mode, with the highest workload, has the lowest error rate on average (7.49%). This indicates that the factory's 6G network has enough capacity.

6.1 Efficiency Distribution by Operation Mode

Operation Mode	High Efficiency	Medium Efficiency	Low Efficiency
Active	2,124 (3.0%)	13,496 (19.3%)	54,434 (77.7%)
Idle	570 (2.8%)	3,792 (18.9%)	15,695 (78.2%)
Maintenance	292 (3.0%)	1,901 (19.2%)	7,696 (77.8%)

Efficiency Class distribution is remarkably equal for all operational modes. The idle mode has marginally less High efficiency (2.8%) and more Low efficiency (78.2%), which can be explained by the fact that machines in idle mode generate errors but no work.

7. Insights and Recommendations

7.1 Insight 1 — Error Rate Control is the Primary Lever

The single most actionable insight that comes from this analysis is that the single most impactful way to improve efficiency classification is error rate reduction. High efficiency. No machine with error rate above 2.0%. On the other hand, no machine with an error rate higher than 5.0% ever achieves High or even Medium efficiency

- Implement real time thresholds for error rate monitoring: alert at 3% and escalate at 5%
- Substitute generic scores with predictive maintenance scoring by error rate
- Root cause: distinguish network-origin errors from mechanical errors via packet loss correlation

7.2 Insight 2 — Production Speed as Efficiency Gateway

High-efficiency machines maintain production speeds above 400 units/hr (mean 450.8), while Low-efficiency machines average only 254.9 units/hr – a 43.4% throughput gap. Speed is partly a result of low error rates (errors cause slowdowns), but also contributes independently to classification.

- Create machine level production speed floor alerts at 380 units/hr as an early degradation signal
- Identify root causes of speed reduction: network latency spikes, mechanical load or operator behaviour
- Measure model speed recovery time after a degradation event – a key latency sensitivity metric

7.3 Insight 3 — 6G Network Operating Within Tolerance

In fact, the fact that latency (1-50 ms) and packet loss (0-5%) do not cause any measurable direct efficiency differentiation is a positive validation of the 6G network infrastructure. • The network error correction and QoS mechanisms are functioning properly in the current operating envelope.

- Sufficient existing 6G infrastructure, no need to upgrade now based on this dataset
- Stress-test the network at higher than 50 ms latency to find the real efficiency collapse point
- Packet loss above 5% (no data in this range in the current dataset)
- Testing 6G network slice for prioritization of latency for heavily loaded machine

7.4 Insight 4 — Systemic Low-Efficiency Baseline Requires Investigation

The Low-efficiency rate of 77.8% is of utmost importance in this data set. It is not an issue of the network, but rather a fundamental operational or configuration issue that needs to be looked into from a machine level, processing level, and workforce management level.

- Machine efficiency profiling for consistent low-performing machines
- Audit the source of the error rate: Do the errors occur in any specific Machine_ID?
- Check shift and time-of-day efficiency variations
- Check the threshold of quality control. Defect rate (mean 5.0%) is same for all machines, implying a possible need for calibration of QC.

7.5 KPI Target Framework

KPI	Current (Avg)	High-Efficiency Target	Priority
Error Rate (%)	7.50%	< 2.00%	Critical
Production Speed (units/hr)	274.9	> 400	High
Network Latency (ms)	25.6 ms	< 30 ms	Monitor
Packet Loss (%)	2.49%	< 2.00%	Monitor
High Efficiency Rate	3.0%	> 25.0%	Strategic
Defect Rate (%)	5.01%	< 3.00%	Medium

8. Conclusion

This work provides a data-driven insight into the correlation between 6G network performance and smart factory efficiency for Thales Group. The study examines 100,000 operational records from 50 machines, in three operation modes, and draws several important conclusions that challenge intuitive assumptions about connectivity-driven manufacturing performance.

First, the 6G network infrastructure is operating reliably in its designed operational envelope. Latency of 1 – 50 ms and packet loss of 0 – 5% do not yield measurable direct differentiation between efficiency classes — showing that the network's QoS and error correction mechanisms work as intended. This is not an indictment of 6G deployment, but a validation.

Second, the key drivers of manufacturing efficiency are operational: error rate control (correlation -0.60) and production throughput (correlation $+0.33$). Network performance affects these metrics in compound failure scenarios but are not network dominated in the current operational window. The 77.8% Low-efficiency baseline is indicative of systemic process, calibration, or workforce challenges that will require a multi-dimensional response beyond investment in the network.

Third, the roadmap to high efficiency is well-defined and quantifiable: machines must achieve simultaneously error rates below 2.0% and production speeds above 400 units/hr. No machine in the dataset reaches High efficiency outside these bounds. This provides a tangible, measurable objective for the Thales operational improvement programme.

Looking forward, the study recommends to extend the network stress-testing envelope beyond 50 ms latency and 5% packet loss in order to characterise the true efficiency collapse threshold — data that would validate 6G investment priorities and guide network slicing configuration for high-criticality machine workloads.

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